



Cloud Native and Transformation





Objectives

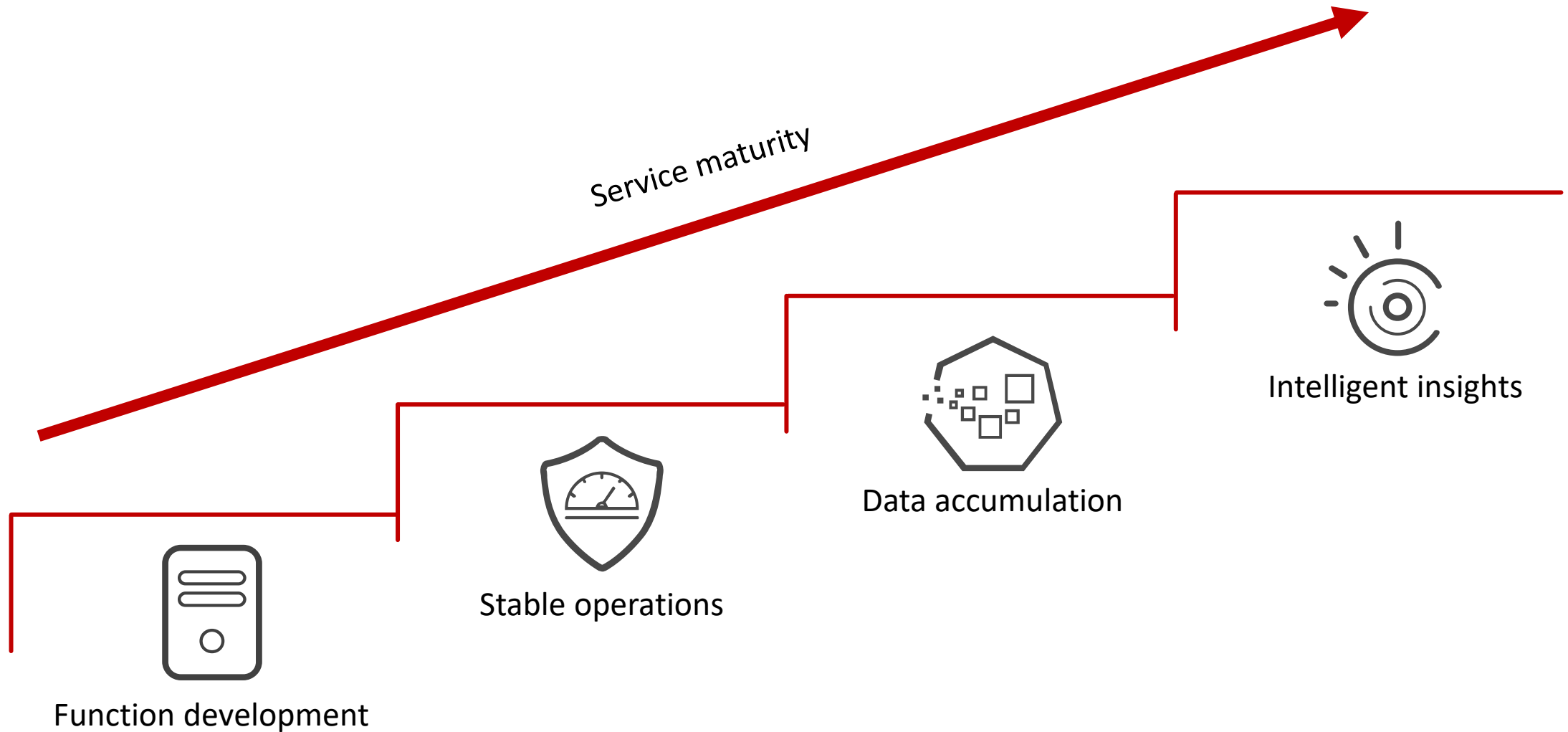
- Upon completion of this lesson, you will:
 - Understand the benefits of cloud native.
 - Understand the key technologies for cloud native transformation.



Contents

- 1. Overview of Cloud Native Transformation**
2. Key Technologies for Cloud Native Transformation

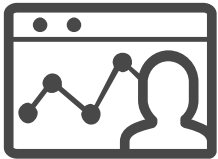
Digital Transformation Promotes Service Maturity



Going Digital with Huawei Cloud



Customer experience



Intelligent operations



Strategic decision making



Technological innovation



Differentiated competitiveness



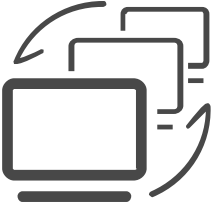
Follow trial-and-error



Focus on the core services



Implement capabilities



Replicate success

Digital transformation goals

How Huawei Cloud helps enterprises transform



Contents

1. Overview of Cloud Native Transformation

2. Key Technologies for Cloud Native Transformation

Cloud Native 2.0 for Better Growth of Enterprises

Efficient resource use

- Higher performance
- Smarter collaboration
- Better lifecycle management

Agile applications

- Secure, trustworthy, efficient, intelligent development
- Flexible governance of dual-stack microservices
- Co-existence of existing and new applications for efficient application integration

Service intelligence

- Intelligent management, more efficient O&M
- Storage-compute decoupling for lower TCO
- One-stop AI platform for inclusive AI

Security & trustworthiness

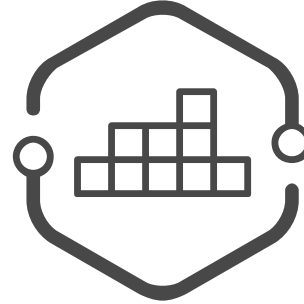
- Consistent security standards for the platform and tenants
- Consistent security standards for 200+ cloud services
- Global compliance

Cloud Native 2.0 Key Technologies



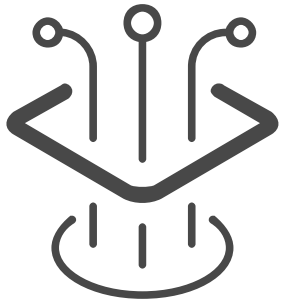
Agile

DevOps



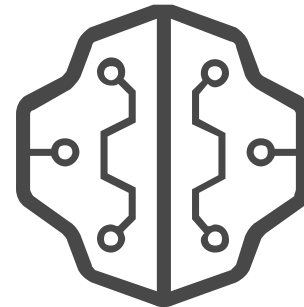
Compute

Containers, functions, and
serverless architecture



Data governance

Data lake and big data



AI

Platforms, components, and
open capabilities

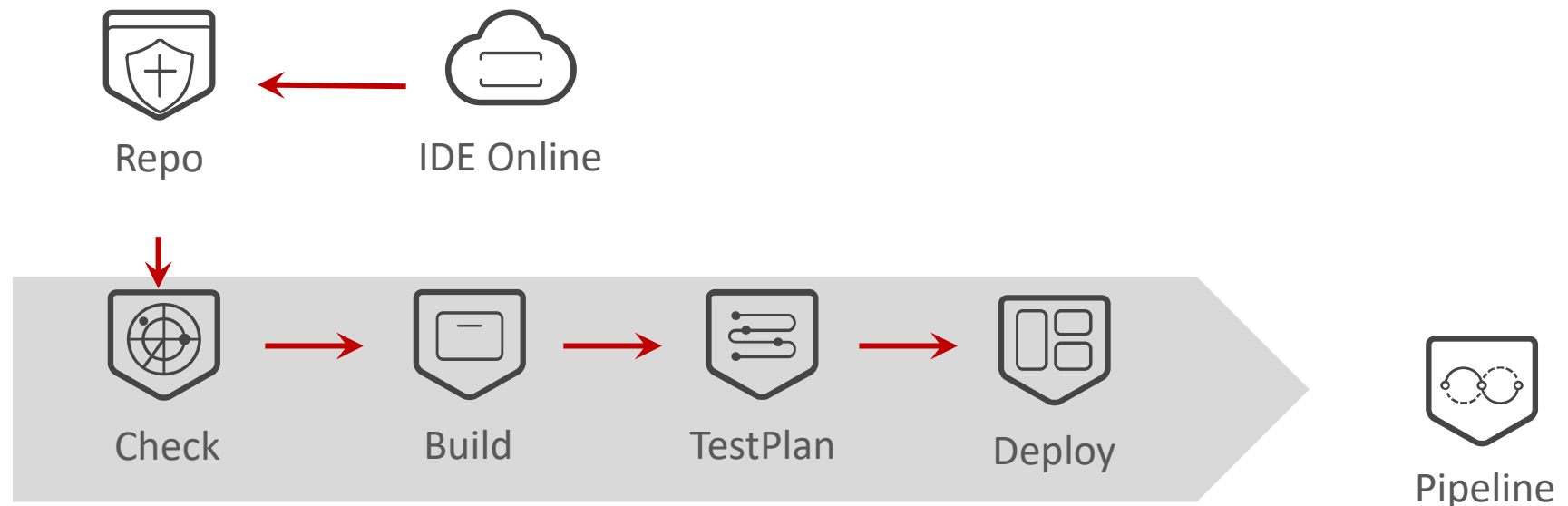


Agile

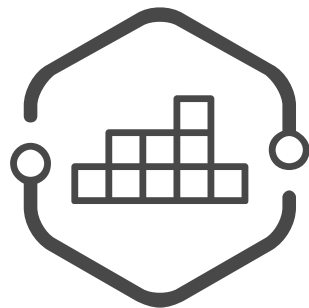
DevOps

Agile Development Keeps Enterprises Innovating and Leading Digitalization

- Huawei Cloud provides a complete agile development system with all required components. Its integrated development environment is easy to use.
- An agile development process means quick go-to-market process and quick response to customer needs.
- Innovation needs the cloud. Agility is the catalyst for innovation.



CodeArts



Compute

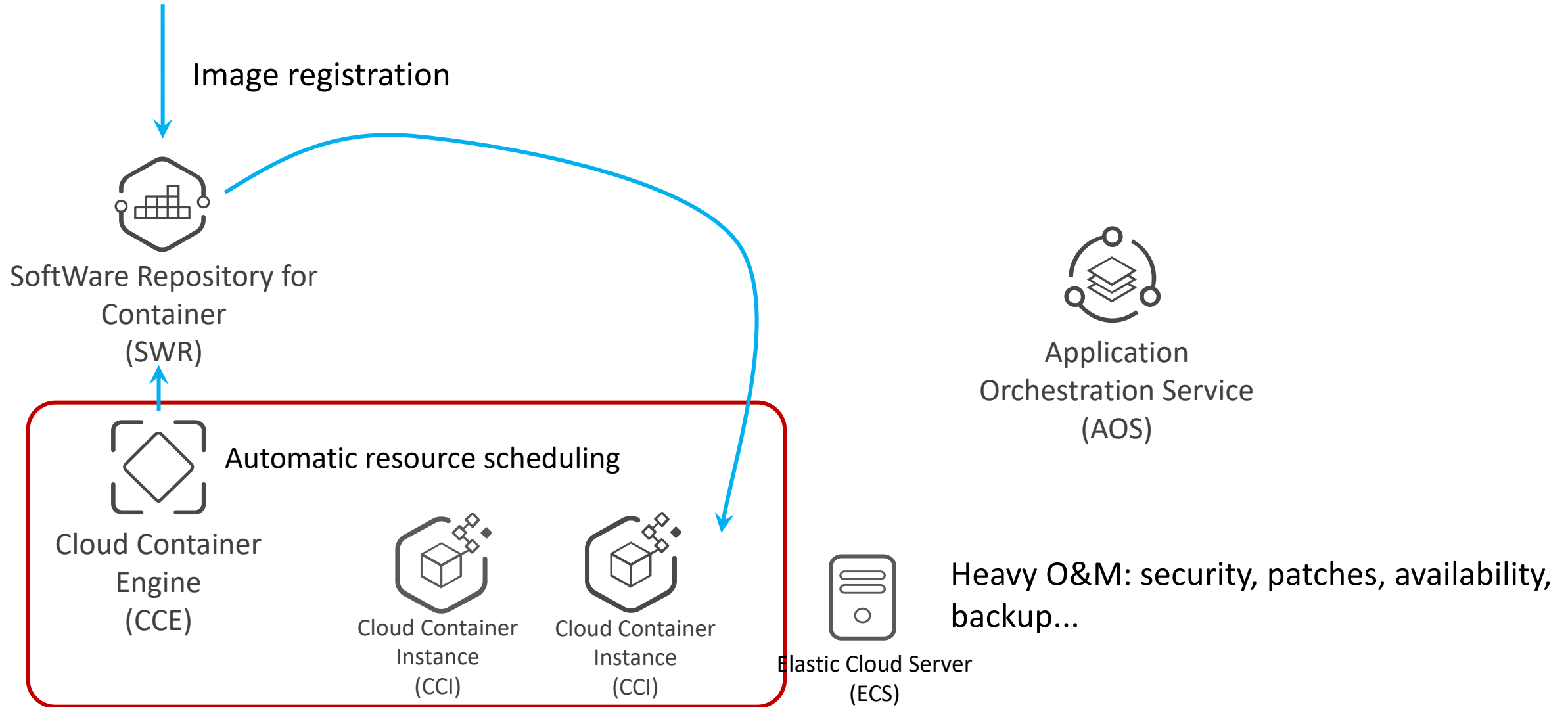
Containers, functions, and
serverless architecture

How Are Servers Related to Customers' Services?

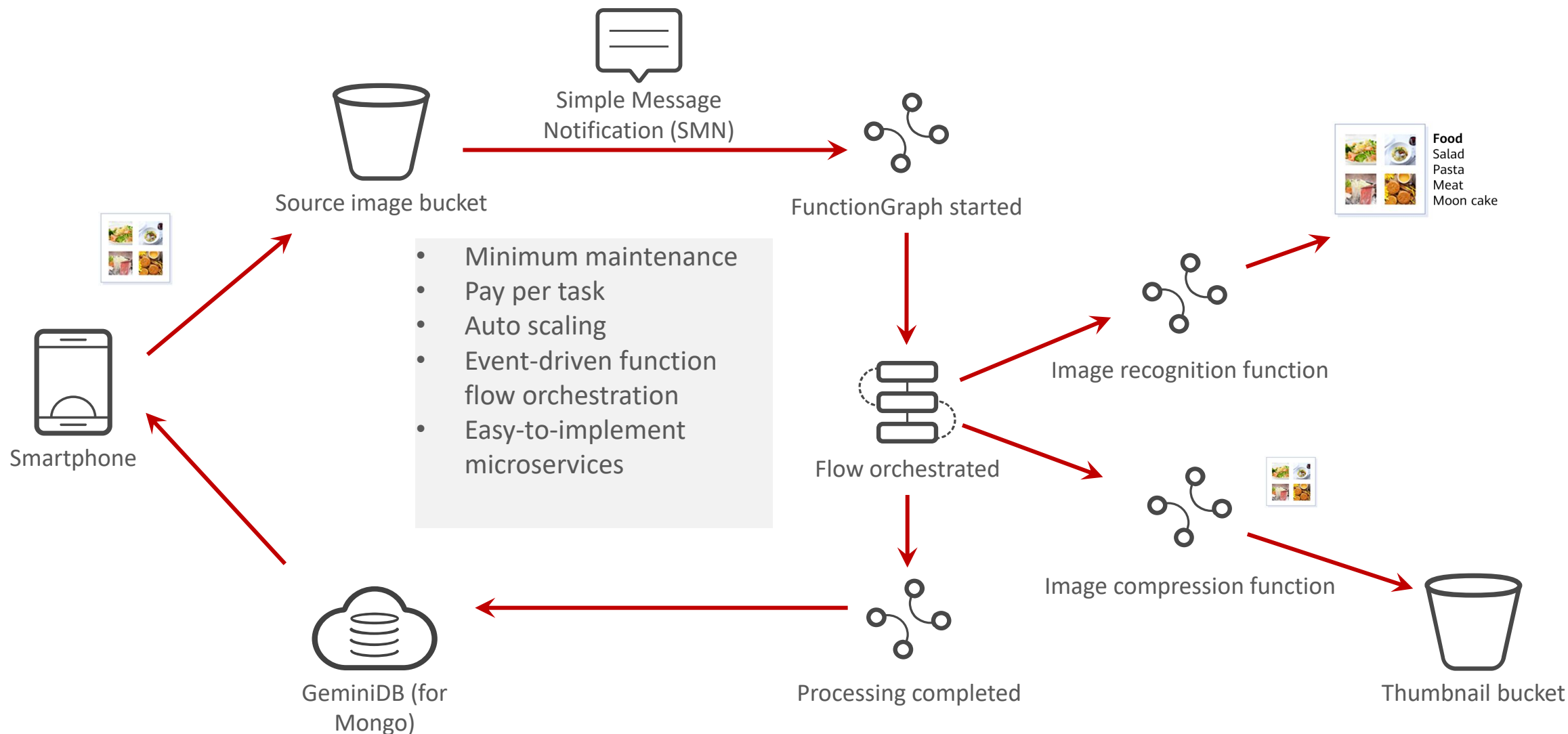
Services are used to run customers' service logic.

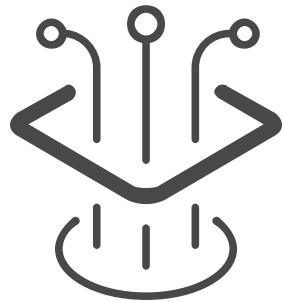
To adapt to changing customer demands, we design services, not servers.

Containers Have Lower Resource Management Overhead



Serverless Architecture: Only Code Logic Needed

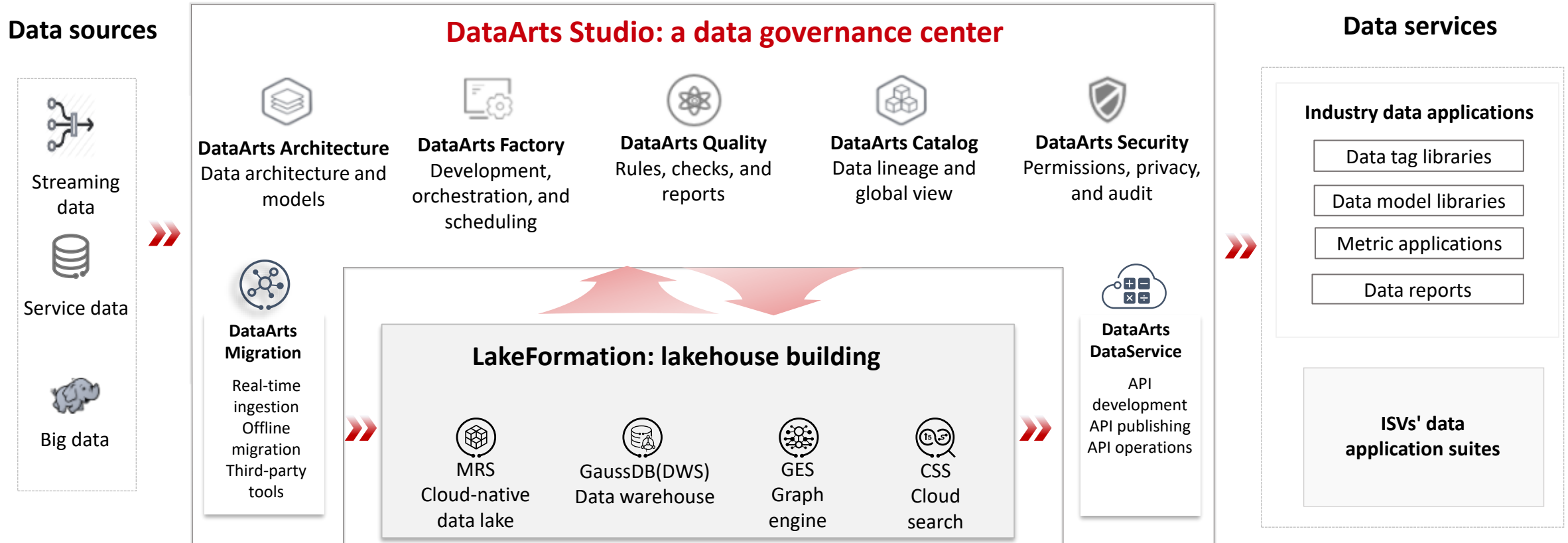




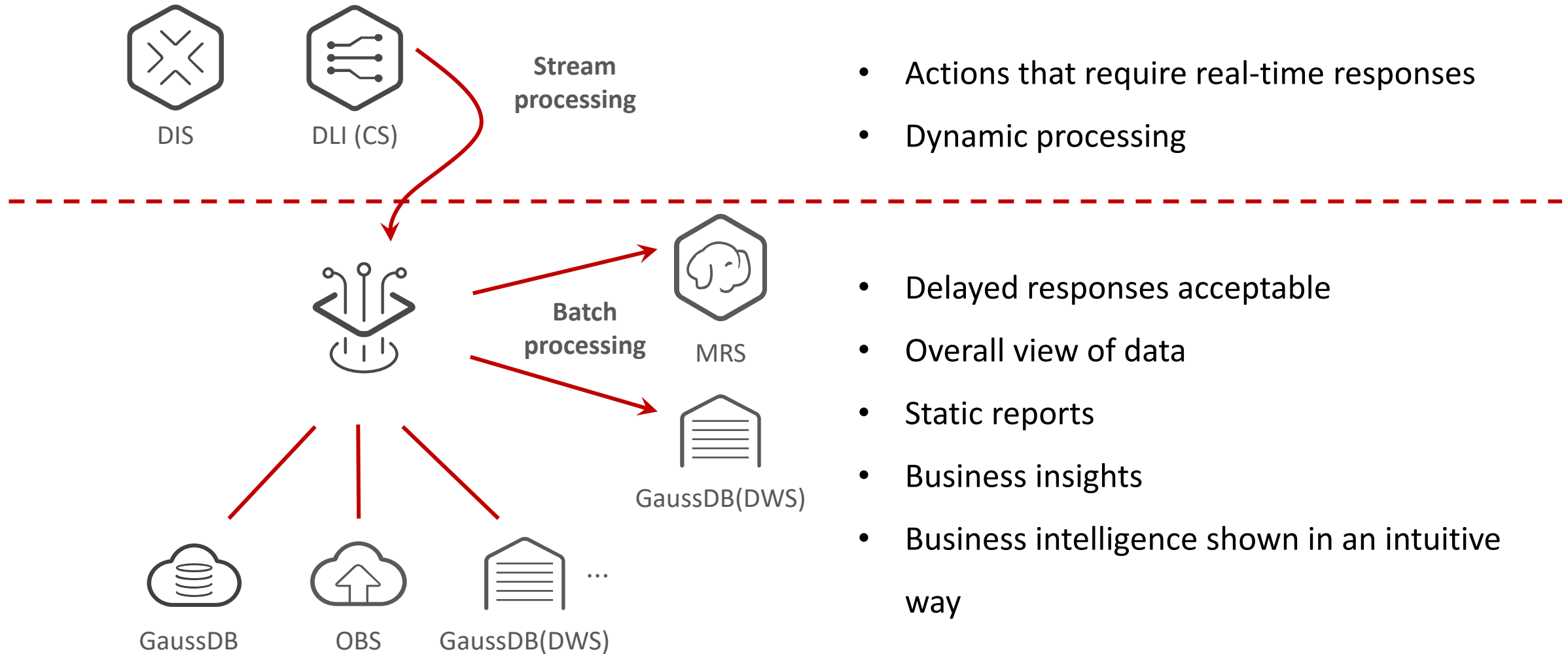
Data governance

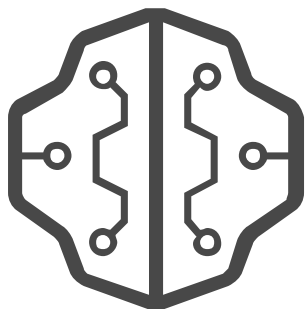
Data lakes and big data

Data Governance Empowers Enterprises to Extract More Value from Data Assets



Big Data: Data Insights Help Unlock Business Potential

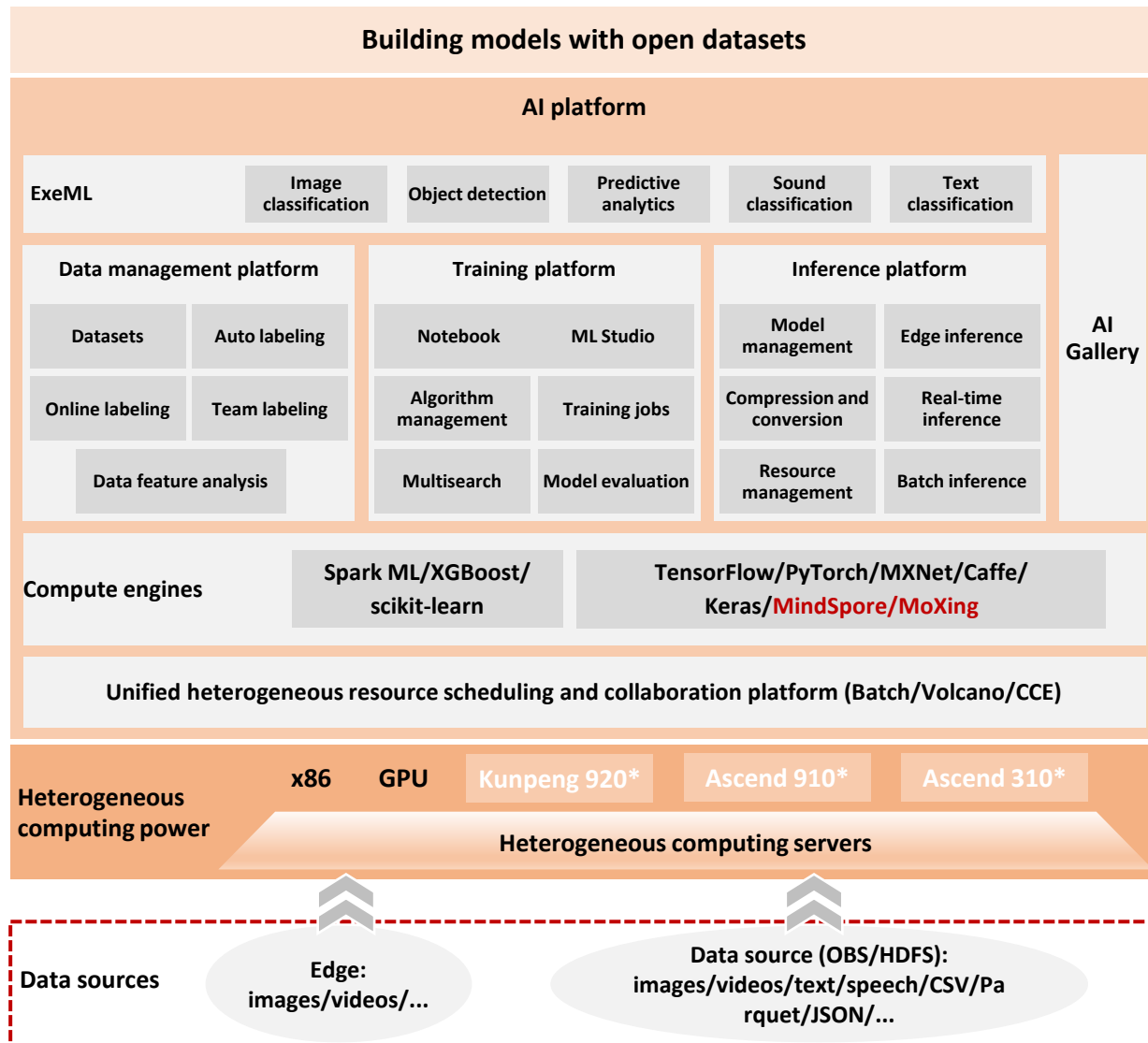




AI

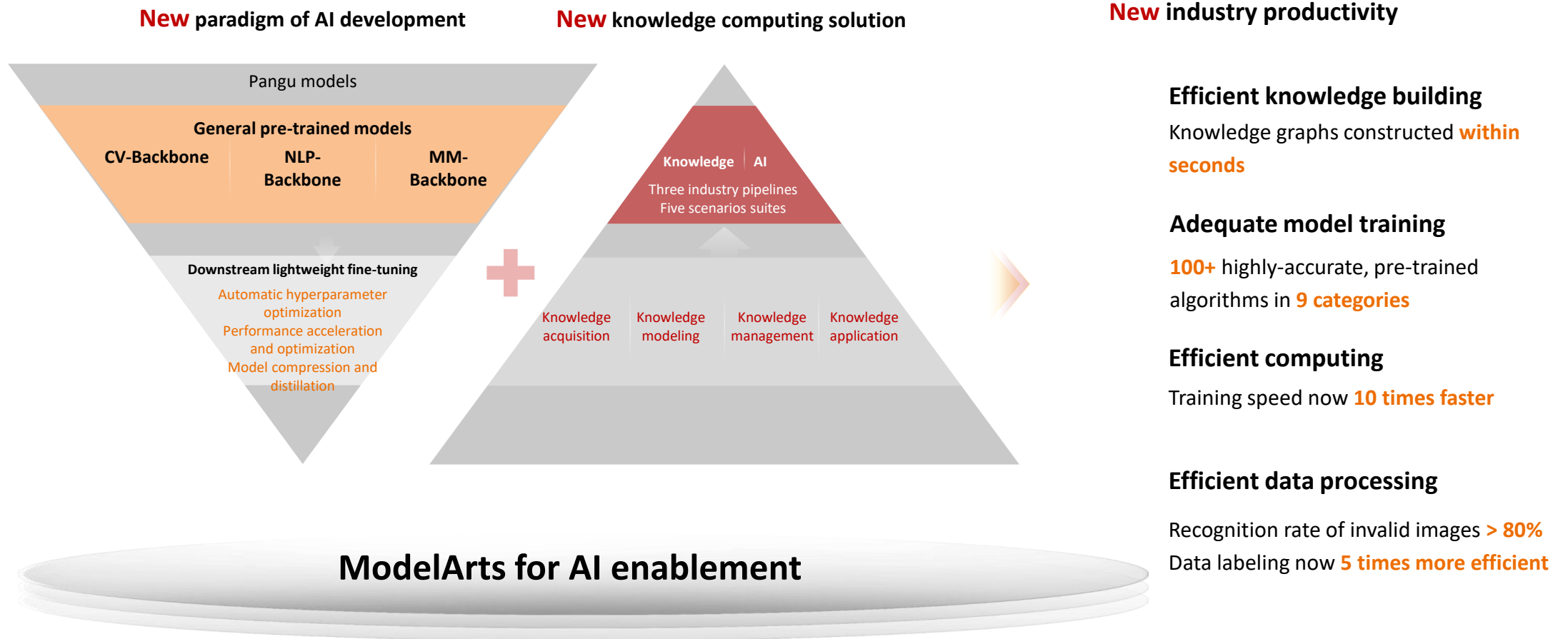
Platforms, components, and
open capabilities

ModelArts: One-Stop AI Development Platform

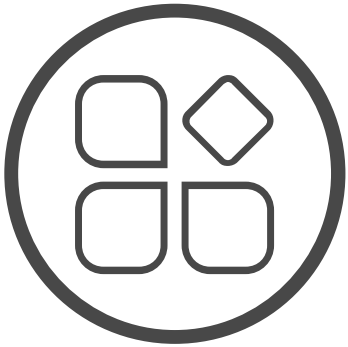


- **Full-stack, full-process, and full-scenario**
 - Software-hardware synergy
 - E2E AI development experience
 - Device-edge-cloud joint development and deployment
- **Unified management and scheduling**
 - Both major (x86 and GPU) and Huawei-developed (Kunpeng and Ascend) chips
 - 10,000-node compute clusters
 - Resource scheduling within seconds and on-demand use
 - 30% higher performance for training jobs
- **Zero-cost migration with the support of major engines**
 - Major engines: TensorFlow, PyTorch, MXNet, and Caffe
 - Huawei-developed engines: MindSpore and MoXing
 - Custom images
- **Multi-dimensional features for customer needs**
 - Beginners: ExeML, pre-trained algorithms, and engineering templates for simplified development
 - Experts: Data augmentation, model diagnosis, and federated learning for more efficient AI development
 - Algorithm developers: In-cloud Notebook and on-premises PyCharm Toolkit

Inclusive AI: Taking Industry AI to New Heights



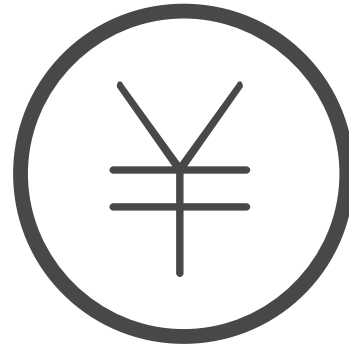
Huawei Cloud Assists in Business Transformation



Comprehensive



Secure & trustworthy



Cost-effective



Easy-to-use



Section Summary

- You have learned about the benefits of cloud native transformation, as well as four key technologies: agile, compute, data governance, and AI.

Quiz

(T/F) In a serverless architecture, developers do not need to configure or manage servers while developing applications. This means that there are no physical or virtual servers in the serverless environment.



Recommendations

- Huawei Cloud websites
 - Huawei Cloud: <https://www.huaweicloud.com/intl/en-us/>
 - Huawei Cloud Developer Institute: <https://edu.huaweicloud.com/intl/en-us/>



Huawei Cloud
Developer Institute

Thank You.

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